

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Short Answer Test

COURSE NAME:	Cloud Computing and Big Data Analytic	COURSE CODE:	CSA1522
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COURSE OUTCOME COVERED

CO1: Apply cloud computing technologies including hardware- software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)	Total Marks: 50
Question Count: 5	

Code of Conduct

I Brijesh.T/(192521391) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____



Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Compare SOAP-based and REST-based web services in terms of communication style, data exchange, and suitability for cloud applications.
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

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Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant example	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.

Answer:

Cloud computing is the delivery of services like storage, servers, software, and databases over the internet instead of using local computers or servers.

Characteristics:

1. On-Demand Self-Service – Users can access resources whenever needed.
2. Broad Network Access – Services can be used from any device with an internet connection.
3. Rapid Scalability – Resources can be increased or decreased easily.
4. Pay-as-You-Go – Users pay only for the services they use.

2. Trace the evolution of cloud computing from hardware evolution to internet software evolution.

Answer:

1. Hardware Evolution – Large mainframe computers were shared by many users.
2. Personal Computers – People started using their own computers.
3. Networking – Computers were connected through LAN and WAN.
4. Internet Evolution – Services became available online.

5. Virtualization – One server could run many virtual machines.
6. Internet Software – Software was accessed through web browsers.
7. Cloud Platforms – AWS, Azure, and Google Cloud started providing online computing services.

3. Explain the role of server virtualization in cloud computing. Differentiate virtualization and cloud computing.

Answer:

Server Virtualization: It divides one physical server into multiple virtual servers, improving resource usage and reducing cost.

Virtualization	Cloud Computing
Creates virtual machines.	Provides services over the internet.
It is a technology.	It is a service model.
Example: VMware	Example: AWS

4. Compare SOAP-based and REST-based web services.

Feature	SOAP	REST
Data Format	XML	JSON, XML
Speed	Slower	Faster
Communication	Protocol-based	Web-based
Cloud Use	Enterprise applications	Cloud and mobile apps

5. Differentiate IaaS, PaaS, SaaS, and XaaS with one example each.

Mode	Description	Example
IaaS	Provides infrastructure like servers and	Amazon EC2

storage.

PaaS	Provides a platform to develop applications.	Google App Engine
SaaS	Provides ready-to-use software online.	Google Docs
XaaS	Provides any IT service through the cloud.	Microsoft Azure AI Services